IoB: Sensors for Wearable Monitoring and Enhancing Health Care Systems

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The health care industry is very rapidly embracing the Internet of Bodies (IoB). The reason for this trend is that IoB features are widely integrated into medical devices that improve efficiency and quality of service. Ongoing surveillance is needed for seniors and patients with chronic diseases. The IoB health care spending solution will reach an astounding US $1 trillion by 2025 for some estimates. It sets the stage for accessible, highly personalized and timely health services for all. The sensors will be integrated into the environment or worn by the user to collect a wealth of information about mental and physical health. Such information, when exploited and aggregated on a continuous and effective basis, can lead to positive changes in the health care landscape. The patient's physiological state is monitored clinically using IoB sensors. By analyzing the collected information, the data will be sent to the processing centers for appropriate action in the cloud environment. The data will check the condition of normal people as well as the sensors. This paper focuses on an IoB application for health care using sensors like KY-039, LM35 and DB1820 and implementing Python middleware for data scripting.

Introduction

Internet of Things (IoT) interfaces with the body and their outcome data comprise the IoB. It is an expansion of the IoT and fundamentally interfaces the humans with an organization through devices that are ingested, embedded, or associated with the body somehow or another. Once associated, the information is traded, and the body and devices can be distantly observed and controlled [1]. IoB may invoke a couple of considerations that have nothing to do with the real essence of the term. However, it is tied in with utilizing the human body as the most recent informational stage. From the start, the idea appears to be very frightening, yet then when you understand the potential outcomes it makes, it turns out to be very energizing. This paper investigates that the IoB and discusses a few models used today and a couple of the difficulties they present. As the IoB advances, numerous individuals are now associated with it through wearable devices. The smart watch developed into a US $13 billion market by 2018 and is projected to build another 32% to US $18 billion by 2021. Smart toothbrushes and even hairbrushes are being considered as instruments to measure individuals' condition and behavior. [2].

For wellbeing professionals, the IoB opens the entryway to another period of powerful observation and treatment. In 2017, the U.S. Government Drug Administration supported the primary utilization of computerized pills in the United States. Advanced pills contain minuscule, ingestible sensors that are consumed just as medication [3]. When taken, the sensor is actuated in the patient's stomach and sends information to their cell phone or different devices. In 2018, Kaiser Permanente, a medical care supplier in California, began a virtual recovery program for patients recuperating from coronary failures. The patients imparted their information to their suppliers for consideration through a smart watch, taking into account better checking and a nearer, more persistent connection between patient and specialist [4].

Internet-associated devices like smart indoor regulators, voice-actuated partners, and web-empowered appliances have become omnipresent in American homes. These advancements are essential for the IoT, which has thrived as of late as customers and organizations are attracted to these devices for accommodation, productivity, and, by and large, fun. IoB innovations fall under the more extensive IoT umbrella. Yet, as the name suggests, IoB devices present a significantly more personal interchange among people and the devices. IoB devices screen the human body, gather measurements and other individual data, and send that information over the web. Numerous devices, like wellness trackers, are now being used [5].

IoT's sheer size of 30 billion (and then some) web-associated devices is amazing, and now we acknowledge that portion of devices that are associated with or at some point inside the human body. This developing industry of devices that screen the human body and send the information gathered using the web is growing each year. IoB alludes to devices put or embedded in the human body that keep a constant exchange between an assortment of contact focuses through the internet. This innovation utilizes the body as a data platform [6].

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Undoubtedly, IoB can permit individuals with ailments, like diabetes, to have better quality and longer life. The innovation permits us to screen blood glucose progressively and control the body's capacities utilizing a portable application. Our physical and virtual universes are mixing and improving comprehension of three key zones: personalities are expanding as they become hyper-improved and multi-tactile; coordinated efforts are associated; and advancements are assorted and comprehensive. It suggested that our bodies have at long last become the interface. IoB is an expansion of the IoT and essentially interfaces the human body to organizations through devices that are ingested, embedded, or associated with the body everywhere [7]. About 70% of users of wearables believe that the manufacturers should be intense about securing their information, especially if they are bound to impart data to specialists, insurance agencies, and web organizations. Even though IoB is considered the future, scientists accept that what is to come is not so distant. As of now, innovations have cleared the way for the medical care to make extraordinary upgrades in the field of information and data checking [8].

Because of aging demographics, in contrast to previous decades, older individuals are progressively getting more isolated in their homes. As individuals get older and infirm, they get defenseless against various clinical and physiological issues. The issue becomes critical when the elderly are alone, and there is no one to deal with their care [9] or help them if any accident occurs. On the off chance that an older person falls and is harmed, it may not be possible to call the rescue vehicle or their family members for help. Rural India comes up short on an office to enlist a guardian or a medical attendant who might assist older individuals living alone. Thus, older people generally conduct day-by-day activities and remain on their own until they die [10].

Wellbeing is consistently a significant worry, and in each development, humanity is progressing as far as innovation. The new COVID-19 assault has significantly affected the world economy and to a degree is a model of how medical services have critical significance. In all regions where the scourge is spread, it is consistently a superior plan to screen patients utilizing tools for distant wellbeing observation, and an IoT-based framework is the current answer for it. A plan for far off patient monitoring engages perceptions of patients outside of standard clinical settings. The center target of this venture is the plan and execution of a smart patient wellbeing global positioning framework that utilizes sensors to follow patient wellbeing and utilizes the web to educate friends and family on the off chance there are any issues of concern [11].

Related Works

In recent years, the business of wearable devices to monitor pregnancies has been growing quickly. The devices go from fetal screens to multi-practical wellbeing assessment instruments, which aided checking, and recording maternal wellbeing pointers, for example, fetal pulse, blood glucose, and circulatory strain in the home. Pregnant ladies and obstetricians are linked together by wearable devices in a phenomenal manner [12]. With the all-inclusive utilization of the IoT innovations, the development of smart systems for maternal health began to raise individuals' interest. The authors propose an IoT platform for smart maternal medical care using wearable devices, distributed computing, and the key advancements. In other applications, the boards offer data from the home to obstetrics offices in emergency clinics. These systems use the IoT and can significantly relieve the responsibility of clinical staffs, improve work proficiency, refer pregnant women to the specialists, and improve the nature of obstetrical treatment [13].

In the current century, the novel coronavirus introduced itself as a genuine danger to the worldwide human populace. Interventions that combine recent innovations like the IoT, dispersed distributed computing, and man-made brainpower, can successfully address the COVID-19 pandemic. From this viewpoint, this section presents different wearable devices as a feature of the medical services framework toward battling the COIVD-19 pandemic [14]. To begin with, this work surveys the distinctive wearable devices and their use by patients, medical care proficient, frontlines, and worldwide residents to battle COVID-19. Consequently, the significant targets of these wearable devices incorporate instrument monitoring, data sharing, and awareness to limit the danger of Covid contamination. Second, the paper summarizes how wearable devices could be implemented to deal with the COVID-19 pandemic [15].

Wearable devices are attracting expanding consideration in both the academic community and industry where data identified with human wellbeing can be measured continuously and along with human-machine associations can empower a change in outlook in the computerized world. For this shift to happen, green and maintainable energy innovations for controlling adaptable wearable devices are a challenge. The authors of [16] surveyed cutting-edge wearable technology and examine their upsides and downsides, basic material science, and general plan/assessment models. Sensor types, materials, manufacturing, power utilization, and techniques are used to test the stretch ability and adaptability of wearable devices. In light of their use in medical care, modern investigation, primary data checking, military, and shopper devices, an incorporated framework design of wearable, adaptable frameworks are introduced. Finally, future directions for the field of wearable technology are illustrated by covering the parts of across-the-board printable wearable devices, fiber and material hardware, and self-controlled mindfulness wearable frameworks [17].

In this time of heterogeneous frameworks, various advancements were incorporated to make application-explicit arrangements. The new upsurge in acknowledgment of innovations, for example, distributed computing and the pervasive internet has made for the IoT [18]. Also, the expanding Internet, infiltrated with the rising utilization of cell phones, has motivated a period of innovation that permits interfacing of actual items and associating them to the Internet for creating applications, filling a wide scope of needs. Ongoing improvements in the field of wearable devices have prompted the making of another portion in IoT, which has been advantageously labeled the Wearable Internet of Things (WIoT) [19].

Wearable devices are the pervasive innovation of the IoT in everyday life. Productive information being collected from different devices like smart garments, sensing wrist wear and clinical wearables that are integrated in the IoT can improve medical care frameworks. The wearable technology market is presently overwhelmed by users to monitor well-being, security, collaboration, activity, identification, wellness, etc. Wearables allow the intermingling of the physical and computerized worlds which consequently carry individuals into the IoT [20]. The popularity of wearable devices is developing dramatically since they change the way users associate with the environment. For example, 74% of individuals accept that the wearable sensors help them in communicating with the devices around them. From this time forward, one out of three cell phone users will wear at least five wearable technology systems. In addition, 60% accept that wearables in the coming five years will be utilized not only to provide wellbeing-related data; they will be utilized to control objects, open entryways, confirm identification, and consumer exchanges. By connecting wearable articles and devices, the IoT will work with inventive administrations in to an enormous arrangement of situations, like home and work computer networks, smart cities, and medical care [21]. It will incorporate new standards for human-to-machine and machine-to-machine communication. These days, wearable devices are considered the sign of IoT, which can be worn on the human body [22].

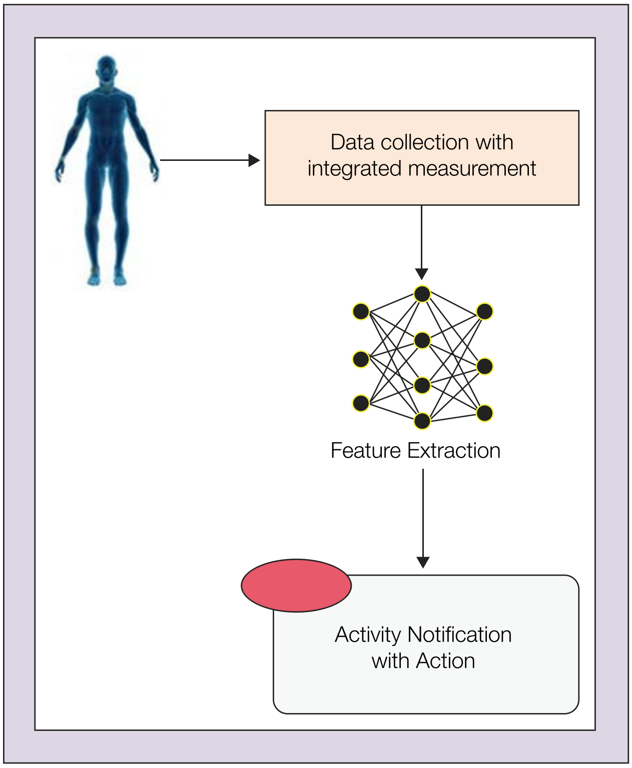

Fig. 1. Internet of Bodies architecture for data transmission.

Methodologies

Data Acquisition

Data acquisition is performed to measure physiological biomarkers by various wearable sensors such as respiratory rate, skin temperature, EMG muscle activity, gait (posture), and ECG. The sensors are connected to a network via an intermediate data concentrator or aggregator that will be in the vicinity of the patient, typically as a Smartphone.

Data Transmission

The system components have the responsibility to convey the patients' recordings from their house to the data center of a Health Care Organization (HCO) with assured privacy and security and in real-time. The system is generally a short-range radio, equipped with a sensory acquisition platform. The components used in the monitoring system for remote patient care are based on the Cloud architecture of IoB. It uses low power such as Bluetooth or Zigbee that transfers the sensor data to the concentrator. To the HCO, the aggregated data is relayed for the long-term storage on a concentrator using internet connectivity through the Cellular data or Wi-Fi connection of a Smartphone. The IoB based architecture is formed by the data acquisition sensors to access the data through the concentrator with internet connectivity. The cloudlet can be direct accessed by the concentrator via a Wi-Fi network which acts as a local processing unit. The cloudlet can be used for aggregated data of patients, for whom there is a need for fast, critical task-related running time. Moreover, the cloudlet can be used to transfer the aggregated data to the cloud when there occur limitations like temporary lack of energy or connectivity in mobile devices (Fig. 1) that may put a patient in danger.

Telemedicine can become more successful by merging wearable devices to make healthcare more accessible and cost-effective, and also increases patient engagement. Wearable App devices can be groundbreaking solutions for issues with health care since they provide continuous monitoring and ubiquitous treatment. Wearable health devices are one of the hottest trends in technology.

Cloud Processing

Cloud processing has three regional components: storage; analytics; and visualization. The design is to store the biomedical information of the patient with long-term storage and assist the professionals by successfully providing health with diagnostic information. In the literature, medical data storage is cloud-based, and the upfront challenges are addressed. Along with sensor data, the e-health records used for health analytics are becoming ubiquitous and can help providers with prognoses and diagnoses for several health diseases and conditions. For such systems, visualization acts as a key requirement when it is non-viable for voluminous data from wearable sensors. Typically, the analyses and data are made in a readily digestible format by visualization are essential when clinical practice is impacted by wearable sensors (Fig. 2).

The LM35 is one sort of regularly used temperature sensors that can quantify temperature with an electrical o/p similar to the temperature (in °C). It can quantify temperature all the more accurately compared with a thermistor. This sensor

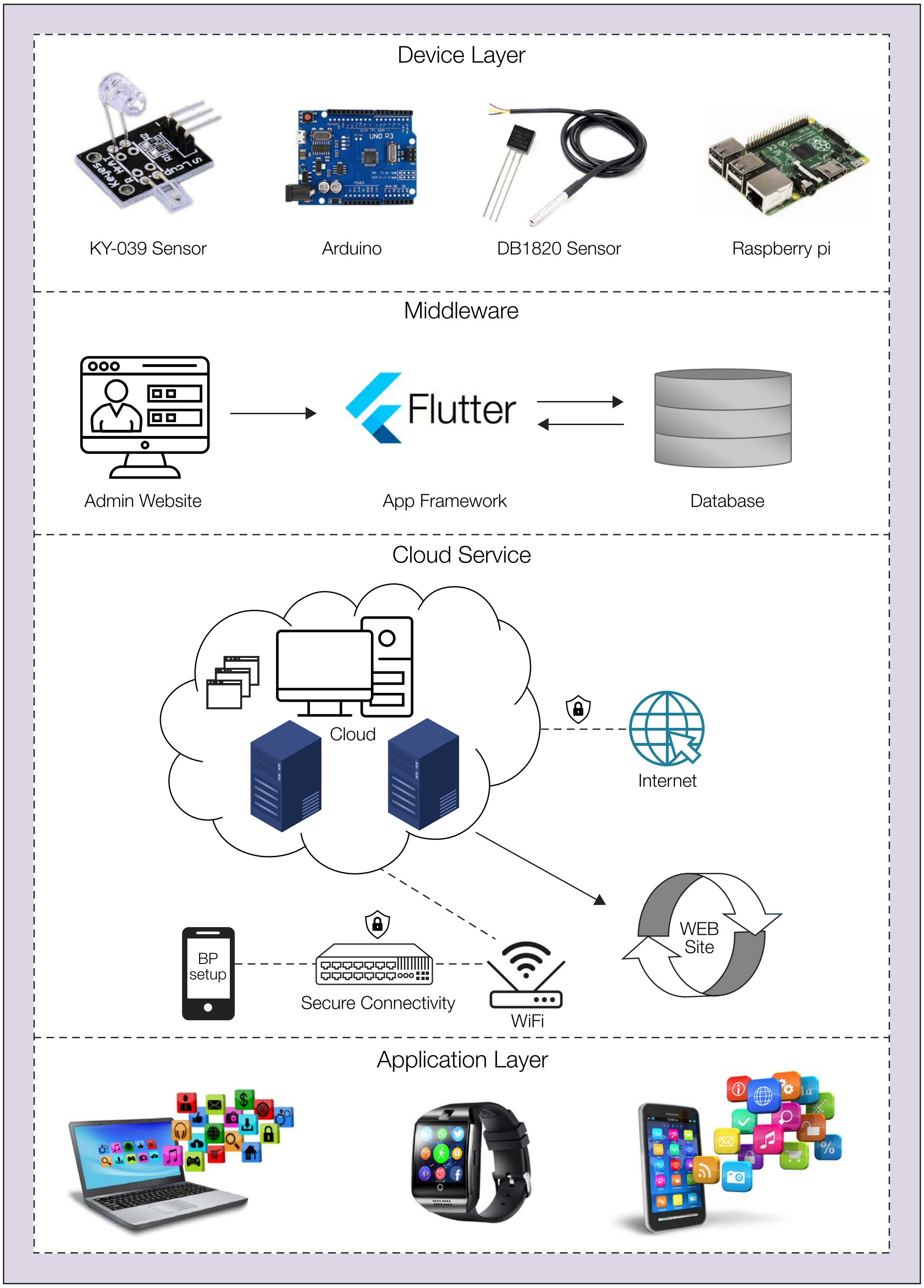

Fig. 2. Graphical abstract for proposed systems.

and detects the time interval in a high pulse. The db 1820 is a highly reactive sensor that can detect the temperature with six digits precision. The temperature of a person is measured using this sensor.

Middleware: The IoB-enabled device with a middleware component helps to make communication between the cloud and device layer. The middleware used for communication monitoring is the Python serial package. The serial communication in the device layer of Arduino transmits the data to the cloud as a request from HTTP that is monitored by Python codes. Through HTTP requests, along with TCP/IP protocol, the messages are sent. This method is very reliable and easily implemented.

Cloud Services: It is open-source, and Math Works, Inc. was chosen for the Application Program Interface (API). The cloud services are much more secure and also provide different platforms like analysis of data and visualization. To read and write, a secured and unique API key provides a security measure.

creates a high yield voltage than thermocouples and may not need that the yield voltage is enhanced. The LM35 has a yield voltage that is corresponding to the Celsius temperature. The scale factor is .01 V/°C.

Device Layer: This layer contains Arduino, Ky-039, and db1820. This layer plays a vital role in the performance reliability of the entire system and ensures that the data that is collected is authenticated. The important function of the layer is data presentation and the decimal point is restricted to 1. Also, the device only has Ky-039 and db1820 sensors. The heart pulse is monitored using the Ky-039 sensor. It emits the Infrared rays

Application Layer: Two freeware Android applications called Bodies Monitor and Thing View may be used to alert medical professionals via the IoB of signals that may be indicative of a patient's acute medical condition.

Implementation

First Phase: The module for initial authentication is built across the paired smart watch and smart phone and eSense. A personal identification number (PIN) will be generated by the user. The PIN will be spoken by the patient through speech-to-text and is processed by the eSense system. The PIN will

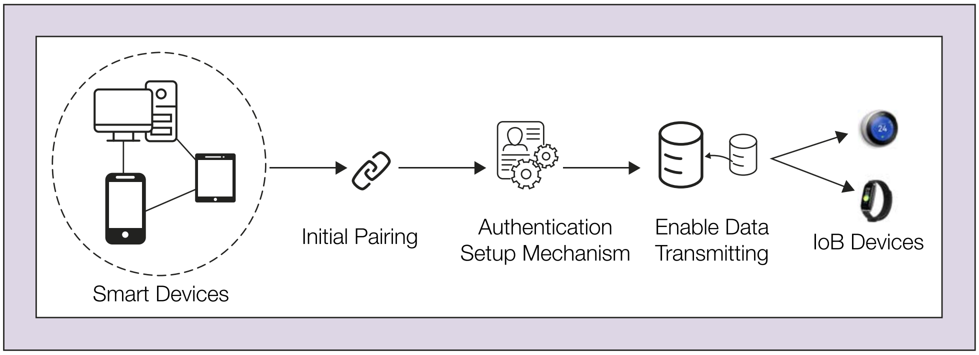

Fig. 3. Initial pairing mechanism with IoB devices.

be received from eSense by the user. Next, the user types the PIN on the watch and then sent to the smart phone. The sensor is enabled in the phone to confirm whether the PIN that is received matches the generated one. The above process of authentication is followed by all the devices to ensure that the sensor array is used by the same person every time.

Second Phase: Time series are used to implicit a secure channel by communication/authentication. The Diffie-Hellman key agreement is used when it is anonymous. In the body area of the network, reviewed methods are introduced for device-to-device authentication and the pairing method is continued. The contextual behaviors are observed to ensure the data collection and accelerometer, velocity, gyroscopic, displacement, and heart rates are extracted uniquely for more devices.

Third Phase: A continuous module of verification is added for smart watches and eSense's same-body monitoring. The contextual modalities like angular velocities, heart rate, and accelerations are collected from two different wearables in time series. The same-body verification features and signatures are extracted from the modalities. The sensor data is windowed to perform the feature analysis. The decision tree is applied to check whether the wearables are on the same body through correlating the modality's features. The data communication will be disabled if the results decision says that the wearable is detached from the body of the user (Fig. 3). One more critical attribute of this sensor is that it draws only 60 micro amps from its stockpile and gains a low self-warming limit.

Measurements Dedicated Software

The firmware called Measurements Dedicated Software is implemented, and it can be categorized into three sections. It can be used to measure the heart rate. Here, the instantaneous heart rate (HRBPM) is measured in beats per minute (BPM), RRInt means RR interval time, and SR means Sampling Rate in the samples per second. The computation occurs in the microcontroller directly, and interval values are buffered as if used to compute the HRV parameters. The waveform of the respiration is buffered from the RAM of PSoC5LP and can be transmitted at every second. The respiration signal is down sampled from the 1000SPS to 6 SPS is the Respiration Signal Processing's first step. An exponential filter is used to smooth the respiration waveform as mentioned in Table 1.

Sensors

The temperature sensor associated with the simple PIN of the Arduino regulator is changed into ADC. Utilizing this computerized information, the regulator changes over it into the genuine temperature in degrees Celsius, utilizing the condition as:

$$\text{Temp} = \left[\text{ADC} * 5/4009 - (390/1000)\right] * (20.5/1000) \quad (1)$$

The heart sensor is dependent on the rule of photo plethysmography. It estimates the adjustment of the blood volume through any organ of the body that causes an adjustment of the luminous power through this organ (a vascular area). Computerized beats are given to a microcontroller to compute the heart rate and BPM given by:

$$\text{BPM} = 60 * f \quad (2)$$

where f is the pulse frequency.

A hygrometer detects measurements and reports both dampness and air temperature. Moistness sensors work by identifying changes that adjust electrical flows or temperature

Table 1 – Arduino specifications

Microcontroller	Atmega 168
Operating Voltage	5V
I/P Voltage	12V
I/O Digital Pins	14
Current Rating	40mA
USB	Mini
Analog Pins	8
SRAM	2KB
Oscillator	16 MHz
EEPROM	1 KB
USART	Yes

Table 2 – Temperature state

Temperature of Body	Stage
36–37 °C	Normal
> 37 °C	High
<36 °C	Low

and are listed in Table 2. The general humidity of Voltage (V) is determined as:

$$V = \left(ADC / 1023\right) * 5 \tag{3}$$

$$\%\ \text{humidity} = \left(V - 0.958\right) / 0.0307 \tag{4}$$

Additionally, the range of heart rates considered to determine the wellbeing of the patient are listed in Table 3. Based on these ranges, the guidelines for diagnosing the illness of the patient are performed. The criteria for several levels of wellbeing are listed in Table 4.

Experimental Results

This paper makes remote monitoring of patients conceivable. As we are focusing more on wearable technology, it is useful for patients as well as caretakers who need to screen their wellbeing status. The advancement here is the utilization of Python as a central product as opposed to utilizing a Wi-Fi module. Python is a flexible language that performs sequential transmission with the Arduino and associates with the think speak server.

This sensor information is then transmitted from the information base worker. This information can be obtained from the cloud by the approved clients utilizing the IoT application stage. The information provided by the sensor about the patient is shown in the application illustrated in Fig. 4.

Fig. 5 and Fig. 6 show the performance analysis of the IoB based sensor system with network delay and the usage of CPU and memory. The sensors were connected to a microcontroller with an integrated 2.4 GHz transceiver. The onboard flash memory allows storing up to 10 h of measurement data. The monitor was implemented as a smart wearables insole that was comfortable to use in regular motor activities and was used to monitor a subject's movement for longer periods of time. However, the insole requires initial calibration for each individual in order to make an inclination correction.

Table 3 – Heart rates	
Heart rate (Pulse Rate)	**Level**
60–100 BPM	Normal
> 100	High
<60	Low

Table 4 –Diagnosing diseases			
Heart Rate	**Temperature of Body**		
	Low	**Normal**	**High**
Low	Checkup	Not healthy	Checkup
Normal	Hypothermia	Healthy	Fever
High	Checkup	Not healthy	Checkup

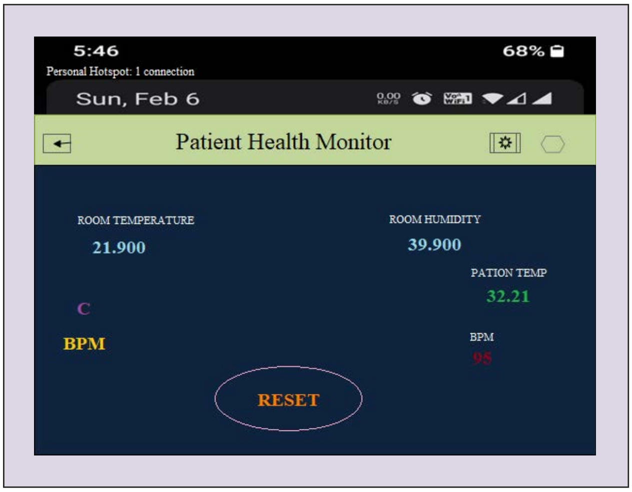

Fig. 4. Sensor-based values displayed on IoT application platform.

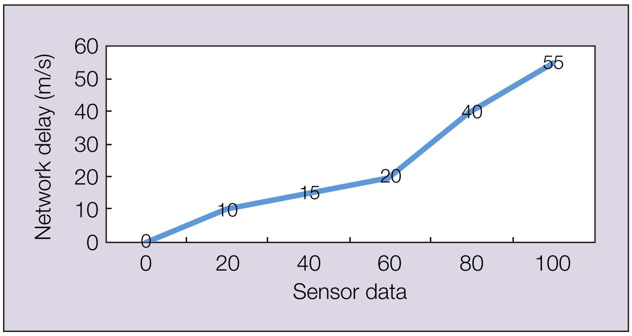

Fig. 5. IoB-based sensor system's network delay.

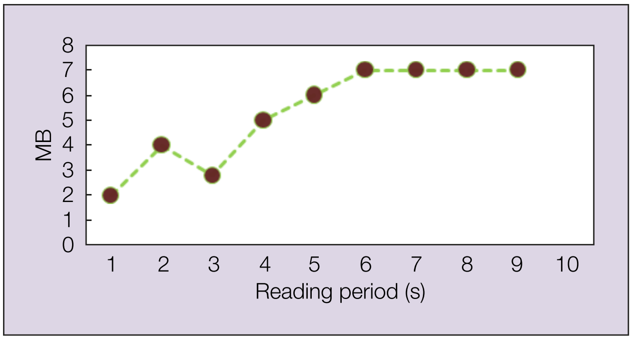

Fig. 6. CPU and memory usage of proposed device

Conclusion and Future Work

There have been numerous applications developed for health care that are used in mobile ambulances and fixed wards of the hospital. The design is accessible easily and flexible where the system only requires to be connected to the network. As it is cloud-based, physical data storage is not required. For doctors, a remote system of monitoring patients will be helpful for effective and timely treatment. The IoB model aims to achieve better treatment by better observing. These frameworks can include different kinds of little physiological sensors and transmission modules. Consequently, the system can work with minimal expense, throughout the day, using wearable sensors that monitor movement and status checking. Through Wi-Fi a notification will be sent to the handset of doctors when there is an increased heart rate and store the data in a cloud environment.

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